

Asymptotic Theory for SCX: Feasibility Analysis

SCX has three growing parameters (K , M , n) and currently only finite-sample (non-asymptotic) concentration bounds. This document assesses whether it is feasible, valuable, and tractable to develop asymptotic theory as these parameters jointly go to infinity.

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1. Executive Summary

Verdict: Asymptotic theory for SCX is feasible in limited regimes ($M \rightarrow \infty$ alone; $n \rightarrow \infty$ alone) but the triple-asymptotic regime ($K, M, n \rightarrow \infty$ simultaneously) is PhD-thesis-level work and not recommended as a near-term goal. The finite-sample bounds already provide stronger guarantees than asymptotic theory would (exponential convergence rates via Hoeffding, covering non-identically-distributed summands via Lindeberg-Feller). For the current paper sequence (AISTATS/TMLR), non-asymptotic bounds are the norm and are preferred. Asymptotic theory could add value at a later stage (e.g., for optimal- K selection rules or phase transition characterization), but should not be a priority.

Bottom Line

Regime	Feasibility	Value	Priority
$M \rightarrow \infty$ (fixed K, n)	Trivial — CLT on $C(x)$	Low (Hoeffding is tighter)	Low
$n \rightarrow \infty$ (fixed K, M)	Easy — consistency of $R_m(s)$	Low (already have PAC bounds)	Low
$K \rightarrow \infty$ (fixed M, n)	Moderate — clustering consistency	Medium (partition recovery)	Medium
$(K, n) \rightarrow \infty$ jointly	Hard — clustering + estimation	Medium-High	Medium

Regime	Feasibility	Value	Priority
(K, M, n) -> oo jointly	PhD-level — triple asymptotics	High but costly	Low (defer)

2. Current Theoretical Landscape

What Already Exists

The SCX framework currently has three theorems and six propositions, all finite-sample (non-asymptotic):

Result	Type	Key Tool	Rate
Thm 1: Noise detection guarantee	$F1 \geq 1 - (1/\eta)^*$ $\exp(-2M \cdot \Delta^2)$	Hoeffding	Exponential in M
Thm 2: Weak feature failure	$AUC \leq AUC_{\text{base}} + O(\sqrt{\delta})$	Pinsker + DPI	$O(\sqrt{\delta})$
Thm 3: Unidentifiability	Construction- based, no rates	Information theory	N/A
Prop 1: Regret lower bound	$\text{Regret} \geq P(\text{bad})^*$ $\delta_{\min}$	Construction	No rate

Result	Type	Key Tool	Rate
Prop 3: State-conditioned weighting	R_SC <= R_global	Jensen/Gibbs	No rate
Prop 4: Compression fidelity		Rademacher	$O(\sqrt{d/n'})$

Key Quantities

The SCX framework has three estimands that could benefit from asymptotic characterization:

1. **State-conditional risk:** $R_m(s) = P(\text{ell}(f_m(X), f^*(X)) < \tau \mid X \text{ in } s)$
 - Currently: PAC bound via Hoeffding: $|R_{\text{hat}} - R| \leq L * \sqrt{\log(2/\delta) / (2 * n_s)}$
2. **Consistency score:** $C(x) = (1/M) * \sum e_m(x)$
 - Currently: Exponential concentration via Hoeffding/Chernoff
3. **State discovery:** k-means on $\phi(X) \rightarrow K$ states
 - Currently: Fano lower bound (Thm 2), no positive result
4. **Noise score:** $NS(x) = r(x) * \rho(s)^{-1} * (1 - C(s))$
 - Currently: no distributional theory for $NS(x)$ itself

Gap Analysis

The existing theory is strong on **concentration** (how fast estimates converge to their means) but weak on: - **Distributional shape:** what does the

sampling distribution look like? - **Joint asymptotics**: how do estimation errors compound across stages? - **Optimal tuning**: what K is information-theoretically optimal? - **Phase transitions**: is there a sharp threshold where SCX stops working?

3. Feasibility by Regime

3.1 $M \rightarrow \infty$ Only (Fixed K, n)

Setup: Fix the input distribution, the state partition, and the number of samples per state. Let the number of experts M grow.

Target: $C(x) = (1/M) * \sum_m e_m(x, y)$.

Current status: Theorem 1 already gives $F1 \geq 1 - (1/\eta) * \exp(-2M*\Delta^2)$, which is an exponential bound. A CLT would give $C(x) \sim N(\mu, \sigma^2/M)$ for large M , but this is a **weaker** guarantee (polynomial in $1/\sqrt{M}$ rather than exponential in M).

Assessment: Feasible but not valuable. The Hoeffding/Chernoff bounds are strictly stronger than any CLT could provide. The CLT would add knowledge about the shape of the distribution (Gaussian around the mean), but the concentration is already known to be exponentially fast.

Difficulty: Trivial. Standard CLT (or Lindeberg-Feller if experts are heterogeneous). The only subtlety is the weak dependence induced by label noise: under (A4), noise samples share a label, creating a common cause. Solution: condition on the noise label, or note that the dependence is vanishing as M grows (each expert's error is independent given x and the noise label).

3.2 $n \rightarrow \infty$ Only (Fixed K, M)

Setup: Fix the number of states K and experts M . Let the number of samples per state n_s grow.

Target: $R_m(s)$, $SCX_m(s)$.

Current status: PAC bounds give $|R_{\hat{}} - R| = O(1/\sqrt{n_s})$. As $n_s \rightarrow \infty$, $R_{\hat{}}(s) \rightarrow R_m(s)$ almost surely by the strong law of large numbers.

What asymptotic theory would add: - CLT for $R_{\hat{}}(s)$: $\sqrt{n_s} * (R_{\hat{}} - R) \rightarrow N(0, \sigma^2(s))$ - This enables confidence intervals for $R_m(s)$ - Delta method for $w_m(x) = \exp(-\alpha * R_{\hat{}}(s))$: the asymptotic distribution of the routing weights

Assessment: Feasible and moderately useful (confidence intervals are practical), but does not add conceptual depth. The existing PAC bounds already cover the consistency story.

Difficulty: Easy. Standard CLT + delta method. The only complication is that n_s grows endogenously (samples fall into states based on clustering), but if we condition on the state partition, n_s is just a random variable with mean $n * P(s)$.

3.3 $K \rightarrow \infty$ Only (Fixed M, n)

Setup: Fix the number of experts M and the total sample size n . Let the number of states K grow.

Target: The state partition itself.

Current status: Theorem 2 gives a lower bound (if features are weak, you can't recover states) but no positive result about when state discovery works.

What asymptotic theory would add: - As K grows, the states become finer, so within-state homogeneity improves - But each state has fewer samples, so within-state estimates become noisier - This is a classic bias-variance tradeoff for state discovery - A consistent partition estimator would prove that as $\phi(x)$ becomes increasingly informative (e.g., $I(\phi; S)$ grows with dimensionality), the estimated partition recovers the true partition

Framework: This is a clustering consistency problem. For k-means: - Pollard (1981) gives $\sqrt{n}$ consistency for centroids in fixed- K setting - For growing K , the problem connects to high-dimensional clustering and rate-distortion theory - The relevant quantity is $I(\phi; S) / (K * \log M)$ — how much information is available per cell

Assessment: Moderately feasible for *consistency* (does the partition converge?), harder for *rates* (how fast?). The key difficulty is that the “true” state partition is itself a modeling choice, not a ground truth uniquely determined by the data distribution.

Difficulty: Hard but tractable. Pollard-style arguments can be extended to growing K under regularity conditions on the feature distribution (e.g., the density is smooth and K grows no faster than $n^{\{1/3\}}$).

3.4 $(M, n) \rightarrow \infty$ Jointly

Setup: Both experts and samples grow simultaneously. The canonical scaling question: what's the optimal ratio between M and n ?

Target: The interaction between two stages of estimation: - $C(x) = (1/M)$

* $\sum e_m(x)$ becomes more reliable as M grows - $R_m(s)$ = average loss in state s becomes more reliable as n_s grows

What asymptotic theory would add: - To achieve SCX reliability estimates with error ϵ , need $M = \Omega(\log(1/\epsilon) / \Delta^2)$ (from Theorem 1) and $n_s = \Omega(K * \log M)$ (from estimation of μ_s) - The joint scaling $M = O(n^\alpha)$ for some α determines which error source dominates - An optimal allocation: given total computational budget ($M * n = B$), what split minimizes the F1 error?

Key technical issue: The quality of the state partition is itself a function of n (more data $\rightarrow$ better clustering). As n grows, the states become better estimated, which tightens Assumption (A5) (state homogeneity), which increases Δ , which reduces the M needed.

Assessment: Harder than each in isolation, but there is a literature on double asymptotics in high-dimensional statistics (Bai & Silverman, random matrix theory) and in semi-supervised learning (the “smoothness” and “cluster” assumptions interact with label propagation).

Difficulty: Substantial. Would need to model the feedback from clustering quality to reliability estimation. The existing Theorem 1 already factors this in implicitly (Δ depends on the state partition quality, which depends on n and ϕ). Making this explicit and deriving joint rates is nontrivial.

3.5 (K, M, n) $\rightarrow \infty$ Jointly

Setup: All three parameters grow. The “full SCX limit.”

Target: - K grows (finer partition) $\rightarrow$ states more homogeneous (A5 tighter)
 $\rightarrow$ Δ increases - But K grows $\rightarrow$ fewer samples per state $\rightarrow$ reliability

estimates noisier - M grows $\rightarrow C(x)$ and μ_s better estimated $\rightarrow$ compensates for noise - n grows $\rightarrow$ states better estimated AND within-state sample size increases

What asymptotic theory would add: - Characterization of the optimal K as a function of M and n - Phase diagram: $(1/\delta)$ - $(1/\rho)$ plane where SCX works - Information-theoretic limits: given $I(\phi; S)$, what's the best achievable F1 in the limit? - Connection to universal consistency of state-conditioned rules

Why this is PhD-level:

1. **Three-parameter asymptotics are rare in the literature.** Even double asymptotics ($K, n \rightarrow \infty$) in clustering is active research. Triple asymptotics ($K, M, n \rightarrow \infty$) with the specific SCX dependency structure is essentially novel.
2. **The state partition is not a fixed object.** As K changes, the “true” state distribution changes. There’s no single limiting object — it’s a sequence of different models indexed by K .
3. **Interacting error sources.** The state discovery error (K dimension), the reliability estimation error (n dimension), and the consistency aggregation error (M dimension) compound in a non-additive way.
4. **Missing mathematical infrastructure.** The standard tools (empirical process theory, random matrix theory, high-dimensional statistics) each handle two-parameter asymptotics but not three. A triple-asymptotic result would likely require novel techniques.

Assessment: The difficulty is genuinely at the level of a 3-5 year PhD project. Not recommended as a near-term goal. However, a **partial** char-

acterization (e.g., phase transition for the special case of binary states and symmetric experts) could be achievable in 6-12 months.

Difficulty: Extremely hard. Would represent a genuine contribution to statistical learning theory.

4. Low-Hanging Fruit: What Can Be Proved with Standard Tools

Despite the difficulty of triple asymptotics, there are several results that can be obtained with off-the-shelf tools and would add value to the SCX theoretical framework.

4.1 Asymptotic Normality of $C(x)$ (Fixed K , n , $M \rightarrow \infty$)

Claim: Under (A1)-(A3), for any fixed x and any threshold τ , as $M \rightarrow \infty$:

$$\sqrt{M} * (C(x) - \mu(x)) \rightarrow N(0, \sigma^2(x))$$

where $\mu(x) = E[e_m(x) | x]$ and $\sigma^2(x) = \text{Var}(e_m(x) | x)$.

Tools: Lindeberg-Feller CLT. The Bernoulli variables e_m are independent given x (by A2) but not identically distributed (different experts have different error probabilities). The Lindeberg condition is satisfied because e_m in $[0,1]$ are bounded.

Value: Low. The Hoeffding bound already gives $P(|C - \mu| > t) \leq 2 \exp(-2 M t^2)$, which is stronger than what the CLT gives (the CLT says

$P(|C - \mu| > t) \sim 2 * \Phi(-\sqrt{M} * t / \sigma)$, which decays as $\exp(-M t^2 / (2 \sigma^2))$ — a similar rate but with a variance-dependent constant in the exponent).

However: The CLT provides the asymptotic distribution, which enables hypothesis testing. For example, testing H_0 : “x is clean” vs H_1 : “x is noisy” can be done using the asymptotic distribution of $C(x)$.

4.2 Confidence Intervals for $SCX_m(s)$ (Fixed K , M , $n \rightarrow \infty$)

Claim: As $n_s \rightarrow \infty$:

$$\sqrt{n_s} * (SCX_hat_m(s) - SCX_m(s)) \rightarrow N(0, SCX_m(s) * (1 - SCX_m(s)))$$

Tools: Standard CLT for binomial proportions. $SCX_hat_m(s)$ is simply the empirical proportion of successes (error $< \tau$) in state s .

Note: The “number of samples in state s ” n_s is itself a random variable (stochastic cluster assignment). But if the state assignment converges as $n \rightarrow \infty$ (see 4.3), the conditional CLT holds.

Value: Moderate. Enables standard error bars on SCX reliability plots. Useful for practitioners.

4.3 Consistency of k-means State Discovery (n , $K \rightarrow \infty$)

Claim: Under suitable regularity conditions on $\phi(X)$ and the true state partition, as $n \rightarrow \infty$ with $K = o(n^{1/3})$:

$$d_H(S_hat, S) \rightarrow 0 \text{ in probability}$$

where d_H is the Hausdorff distance between the estimated and true cluster centers, or the misclassification rate if states are well-separated.

Tools: - Pollard (1981, 1982) for k-means consistency with fixed K - Extensions to growing K : when K grows, need the “excess risk” of k-means to vanish, which requires $K * \log K / n \rightarrow 0$ (for well-separated clusters) or stronger separation conditions - For the SCX setting, the relevant metric is the partition error rate, $P(s(x) \neq \hat{s}(x))$. This is bounded via Fano (Thm 2) but a positive consistency result would write:

$$P(s \neq \hat{s}) \rightarrow 0 \text{ when } I(\phi; S) / \log K \rightarrow \infty \text{ (i.e., } \delta \ll \log K)$$

Value: High. This would be the first *positive* result about state discovery in SCX (Thm 2 only gives a negative result). It would answer: when does the SCX state discovery procedure actually work?

Difficulty: Moderate for well-separated states (standard cluster consistency). Hard for overlapping states (the typical SCX case).

4.4 Delta Method for Expert Routing Weights

Claim: As $n_s \rightarrow \infty$ for all s , the empirical routing weights:

$$\hat{w}_m(x) = \exp(-\alpha * \sum_s \gamma_s(x) * \hat{R}_m(s))$$

converge in distribution to a log-normal limit (approximately).

Tools: Delta method applied to the softmax function. If $\sqrt{n} * (\hat{R} - R) \rightarrow N(0, \Sigma)$, then:

$$\sqrt{n} * (\hat{w} - w) \rightarrow N(0, J * \Sigma * J^T)$$

where J is the Jacobian of the softmax transformation.

Value: Moderate. Provides asymptotic confidence intervals for routing weights. Useful for understanding routing uncertainty.

4.5 The Optimal K: Bias-Variance Tradeoff

This is the most valuable “low-hanging fruit” — a precise characterization of the optimal number of states for SCX.

Setup: As K increases: - Within-state homogeneity improves: $\sup_{x \in s} |P(Y|X=x) - P(Y|X \in s)|$ decreases $\rightarrow \mu_s$ decreases $\rightarrow \Delta$ increases $\rightarrow$ F1 improves - But sample size per state decreases: $n_s \rightarrow n/K$ $\rightarrow$ estimation variance increases $\rightarrow \hat{R}_m(s)$ becomes noisier

Claim: Under regularity conditions (Lipschitz conditional distribution $P(Y|X=x)$ in x), the SCX F1 is optimized at:

$$K^* = (n * c / \log M)^{1/3}$$

for some constant c depending on the smoothness of $P(Y|X=x)$ and the geometry of the feature space.

Derivation sketch: - $F1 \geq 1 - (1/\eta) * \exp(-2M * \Delta^2)$ - $\Delta = (1/2) * (1 - \mu * K/(K-1)) \approx (1/2) * (1 - \mu)$ for large K - μ_s = average expert error rate in state s - Bias term: $\mu_s = \mu_{\text{global}} + O(1/K)$ (as K grows, states homogenize) $\rightarrow \mu_s \leq c_{\text{bias}} / K \rightarrow \Delta \geq 1/2 - c_{\text{bias}}/(2K)$ - Variance term: estimation error in μ_s scales as $\text{Var}(\hat{\mu}_s) = O(1/n_s) = O(K/n)$ - Worst-case $\Delta = 1/2 - c_{\text{bias}}/(2K) - c_{\text{var}} * \sqrt{K/n}$ - Optimizing over K : $d/dK [c_{\text{bias}}/K + c_{\text{var}} * \sqrt{K/n}] = 0 \rightarrow -c_{\text{bias}}/K^2 + c_{\text{var}} / (2 * \sqrt{K * n}) = 0 \rightarrow K^* = (2 * c_{\text{bias}} / c_{\text{var}})^{2/3} * n^{1/3}$

Imposing the M requirement: For the F1 guarantee to be tight, we also

need $M > 1/(2\Delta^2) \log(2/(\eta\epsilon))$. With $K = O(n^{1/3})$, $\Delta = \Omega(1)$, the M requirement is asymptotically constant (does not grow with n).

Value: Very high. This would give practitioners a principled way to choose K given n and M . It would be a practically useful theorem.

5. Hard Problems: What Requires New Theory

5.1 Joint Distribution of the SCX Score

Problem: Characterize the joint asymptotic distribution of $(\text{SCX}_1(s), \dots, \text{SCX}_M(s), C(x), \text{NS}(x))$ as all parameters grow.

Why it's hard: The SCX score is a two-stage estimator: 1. First stage: estimate the state partition (clustering) 2. Second stage: estimate within-state reliability (averaging)

These stages share data (the same n samples are used for both), creating complex dependence that standard asymptotics cannot handle. The state partition is a function of the full data, making all subsequent estimates conditionally dependent even after conditioning on the true states.

Known results in related areas: - “Post-clustering inference” (Lockhart et al., 2014; Lee et al., 2016): selective inference after model selection. But this is for regression coefficients after LASSO, not for plug-in estimates after clustering. - “Double dipping” (Kriegeskorte et al., 2009): a known problem in neuroscience where the same data is used for selection and estimation.

Difficulty: PhD thesis level. Would require novel selective inference tools for clustering-based two-stage estimation.

5.2 Optimal Scaling Rates for Triple Asymptotics

Problem: Find the optimal scaling relationship between K , M , and n such that SCX achieves consistency ($F1 \rightarrow 1$) at the fastest possible rate.

The information-theoretic question: Given: - A fixed total sample budget $T = M * n$ (total training samples across all experts) - A fixed feature space dimensionality d - A data distribution P with true state structure

What scaling $K = f(T, d)$, $M = g(T, d)$, $n = h(T, d)$ minimizes the asymptotic F1 error?

Why it's hard: 1. Three-way tradeoff: M controls detection power, n controls estimation precision, K controls partitioning granularity 2. All three consume resources: more experts means fewer samples per expert (if total data fixed); more states means fewer samples per state 3. The optimal scaling likely depends on the smoothness of $P(Y|X)$ and the clusterability of $\phi(X)$ 4. This is a constrained optimization problem over the space of $(K, M, n) \rightarrow F1$, where $F1$ is a complex function of all three

Connection to known results: - In high-dimensional classification, the optimal rate involves $n / (d * \log p)$ scaling (Fan & Fan, 2008). This is a two-parameter problem. - In clustering, the optimal K scales as $n^{\hat{d}/(d+2)}$ for density-based clustering (the “silverman” bandwidth rule). But SCX is not density-based. - The SCX triple is structurally more complex than either of these.

Difficulty: Open research problem. Would be a significant theoretical con-

tribution if solved.

5.3 Asymptotic Distribution under Model Misspecification

Problem: What happens to SCX when the state partition model is wrong? Specifically, when the true distribution $P(Y|X)$ does not factor through a finite-state partition.

The issue: SCX assumes $P(Y|X) = P(Y|\phi(X))$ where $\phi(X)$ is discrete with K states. If this assumption is false (e.g., Y depends on X in a continuous way that no finite partition can capture), what does the SCX estimator converge to?

What happens in the limit: - As $K \rightarrow \infty$, any continuous function can be approximated arbitrarily well by a step function (the partition becomes a Riemann sum approximation). So SCX should be “universally consistent” as $K \rightarrow \infty$ with $n \rightarrow \infty$ (provided K grows at the right rate). - But as $K \rightarrow \infty$ with fixed n , the within-state estimates become inconsistent (no samples per state).

Why it’s hard: Proving universal consistency of SCX requires: 1. Showing that the excess risk due to state discretization vanishes as $K \rightarrow \infty$ (approximation error) 2. Showing that within-state estimation error vanishes as $n \rightarrow \infty$ (estimation error) 3. Balancing K and n to minimize the sum

This is a **nonparametric regression with state discretization** problem. The rate would be analogous to nonparametric rate for partitioning estimators, but with the additional complexity of the multi-expert aggregation.

Difficulty: Hard but tractable. Follows the standard sieve/penalized M-estimation framework (Grenander, 1981; Shen & Wong, 1994).

5.4 Phase Transition in the (δ, η, K) Space

Problem: Characterize the sharp threshold where SCX goes from “works” to “fails” as a function of feature-information δ , noise rate η , and number of states K .

Conjecture: There is a sharp phase transition in the (δ, η) plane, separated by a curve $\delta_c(\eta, K)$. Below this curve, $F1 \rightarrow F1_{\text{base}}$ (SCX does not help). Above it, $F1 \rightarrow 1$ (SCX works perfectly).

Why it’s hard: 1. Phase transitions in statistical learning are rare (e.g., the “all-or-nothing” phenomenon in compressed sensing). Most learning problems have smooth transitions. 2. The SCX phase transition (if it exists) is in the F1 score, which is a discrete metric (it involves thresholded decisions). Continuous metrics like AUC may not show sharp transitions. 3. Proving a phase transition requires: (a) existence of a threshold where behavior changes, (b) showing the transition is sharp (gap vanishes in the limit), (c) characterizing the threshold location. 4. The triple-parameter space makes this even harder.

Difficulty: Extremely hard for rigorous proof. Moderate for heuristic or physics-style analysis (e.g., replica method or cavity method).

6. Phase Transition Conjectures

Despite the difficulty of rigorous proof, we can conjecture the phase diagram based on the existing finite-sample bounds.

6.1 The (K, M, n) Phase Diagram

SCX “works” ($F1 \rightarrow 1$ asymptotically) in the following regime:

Condition 1 (State recovery): $I(\phi; S) / \log K \rightarrow \infty$ (features carry enough information to resolve K states)

Condition 2 (Detection power): $M * \Delta^2 / \log M \rightarrow \infty$ (enough experts to separate clean from noisy)

Condition 3 (Estimation): $n * P_{\min} / (K * \log M) \rightarrow \infty$ (enough samples to estimate within-state rates)

where $P_{\min} = \min_s P(s)$ is the minimum state probability, and Δ is the mean separation gap from Theorem 1.

If any of these conditions fail, SCX degrades in a specific way: - Condition 1 fails ($\Delta \gg \log K$): states are unrecoverable, SCX noise detection degenerates to loss baseline (Thm 2 regime) - Condition 2 fails ($M * \Delta^2$ small): $C(x)$ is too noisy to separate clean from noisy; $F1$ stuck below some constant < 1 - Condition 3 fails ($n * P_{\min}$ small): within-state estimates are pure noise; state-conditioned weighting becomes worse than global weighting

6.2 The Sharpness Conjecture

The most likely candidate for a sharp phase transition is in **Condition 2** (the detection phase transition), because:

1. Theorem 1 gives $F1 \geq 1 - (1/\eta) * \exp(-2M * \Delta^2)$
2. This shows that $F1 \rightarrow 1$ **exponentially fast** when $M * \Delta^2 \rightarrow \infty$

3. Conversely, when $M * \Delta^2 \rightarrow 0$, the bound becomes trivial ($F1 \geq -\infty$)
4. But is there a converse? If $M * \Delta^2$ is small, is $F1$ bounded away from 1? A minimax lower bound would confirm a phase transition.

Conjectured minimax lower bound: There exists a data distribution satisfying (A1)-(A6) such that, for any detection algorithm:

$$E[F1] \leq 1 - c * \exp(-C * M * \Delta^2)$$

for some constants $c, C > 0$. This would show that the exponential rate in Theorem 1 is tight (up to constants) and that $M * \Delta^2$ is the correct phase transition parameter.

Proof approach: Construct a Le Cam two-point testing problem where the null is “all samples clean” and the alternative is “eta-fraction noisy.” The minimax risk of testing these two hypotheses is $\exp(-M * KL(C_{\text{clean}} || C_{\text{noisy}}))$ which scales as $\exp(-M * \Delta^2)$.

6.3 The Triple Phase Transition

The full phase transition (K, M, n jointly) would have the form:

- Phase 1 (SCX works): $K * \log(K) / n \ll 1$ AND $M * \Delta^2 / \log M \gg 1$
- Phase 2 (State failure): $K * \log K / n \gg 1$ (features too weak or K too large for given n)
- Phase 3 (Detection failure): $M * \Delta^2 / \log M \ll 1$ (too few experts)
- Phase 4 (Double failure): Both Phase 2 and Phase 3 hold

In Phase 2, SCX noise detection works no better than the loss baseline (Thm 2 applies). In Phase 3, the detection guarantee fails but the state discovery

may still be valid (SCX can still be useful for expert routing, just not for noise detection). In Phase 4, SCX entirely fails for all tasks.

7. Would Reviewers Care?

7.1 AISTATS / TMLR / NeurIPS Norms

Current norm: Non-asymptotic (finite-sample) bounds are standard and preferred in top ML venues. The majority of learning theory papers at NeurIPS, ICML, COLT, and AISTATS use PAC-style bounds.

Why finite-sample is preferred: 1. **Non-asymptotic bounds are stronger:** they hold for any finite n, M, K , not just “in the limit.” A reviewer who sees both a CLT (asymptotic) and a Hoeffding bound (finite-sample) will correctly note that the Hoeffding bound is strictly more informative for practical sample sizes. 2. **Rate-sharpness is the focus:** what matters is how fast the error decays in $n/M/K$, not the limiting distribution. The exponential rate in M from Theorem 1 is more impactful than a CLT would be. 3. **Asymptotic arguments can hide complexity:** Dependencies that matter at finite sample sizes wash out in the limit. Reviewers are suspicious of asymptotic arguments that ignore these dependencies.

7.2 When Asymptotics Add Value

There are specific scenarios where asymptotic results strengthen a paper:

1. **Minimax lower bounds:** These are often asymptotic ($n \rightarrow \infty$ for fixed d , or $d \rightarrow \infty$ with n) but provide fundamental limits. A minimax

lower bound for SCX (e.g., “no algorithm can achieve $F1 > \text{constant}$ unless $M * \Delta^2 > \text{threshold}$ ”) would be a strong contribution.

2. **Phase transitions:** The compressed sensing literature (Donoho, Candes, Tao) was hugely influential precisely because it characterized a sharp phase transition. If SCX has a provable phase transition, it would be a major selling point.
3. **Optimal scaling:** A result like “the optimal K scales as $K^* \sim (n / \log M)^{1/3}$ ” would be practically useful and theoretically interesting. It would show that the theory has actionable implications.
4. **Asymptotic equivalence:** Showing that SCX is asymptotically equivalent to some well-studied procedure (e.g., the Bayes classifier for the state-conditional model) would legitimize the approach.

7.3 Recommended Position for Current Papers

For the first SCX paper (primary submission): - Keep the current finite-sample bounds (Theorems 1-3) - Add the following asymptotic *conjectures* as open problems in the discussion section (not as main results): - “The optimal number of states K^* is conjectured to scale as $O(n^{1/3})$ ” - “A phase transition in $M * \Delta^2$ is conjectured” - This shows the authors are aware of the asymptotic questions without requiring full proofs

For the second SCX paper (follow-up): - One clean asymptotic result (e.g., the optimal K scaling or a minimax lower bound) would strengthen the theoretical contribution - By then, the framework will be established, and the asymptotic question will be a natural next step

7.4 Review Criteria

What a reviewer would look for in asymptotic theory:

Criterion	What they want	How SCX scores
Novelty	Is this a new result or an application of an existing technique?	Moderate-high (unique triple structure)
Cleanliness	Is the result clean and interpretable?	Potentially high (phase transitions are inherently clean)
Assumptions	Are assumptions justified?	Tricky (asymptotics require stronger assumptions)
Practicality	Does the result help practitioners?	High if optimal K is derived
Technical depth	Is the proof nontrivial?	Would be, for triple asymptotics

8. Estimated Effort

Feasibility Matrix: Effort vs. Impact

Result	Effort	Impact	Ratio	Priority
CLT for $C(x)$ ($M \rightarrow \infty$)	1-2 days	Low	Low	4
CI for $SCX_m(s)$ ($n \rightarrow \infty$)	2-3 days	Low- Moderate	Medium	3
k-means consistency ($n, K \rightarrow \infty$)	2-4 weeks	High	High	1
Optimal K scaling	2-4 weeks	Very high	Very high	1
Delta method for routing weights	1 week	Low- Moderate	Low	4
Minimax lower bound (M vs $F1$)	2-4 weeks	High	High	2
Joint (M, n) asymptotics	1-3 months	Medium- High	Medium	5
Triple (K, M, n) asymptotics	3-12 months	Very high	Low (cost)	6
Phase transition (full)	3-12 months	Very high	Low (cost)	6

Result	Effort	Impact	Ratio	Priority
Post-clustering inference for SCX	3-6 months	High	Medium	5
Universal consistency (nonparametric)	2-6 months	High	Medium	5

Recommended Priority Order

Phase 1 (before first paper submission, 1-2 months): 1. Cluster consistency: prove that k-means on $\phi(X)$ recovers the true partition as $n, K \rightarrow \infty$, under appropriate conditions (informally: “the SCX state discovery procedure is consistent”) 2. Optimal K scaling: derive $K^* = (n / \log M)^{1/\text{dimension}}$ from the bias-variance tradeoff (informally: “K should grow slowly with n”)

Phase 2 (between submission and revision, 1-2 months): 3. Minimax lower bound for F1: show that no detector can exceed $F1 > 1 - c * \exp(-C * M * \Delta^2)$, proving Theorem 1 optimal up to constants (informally: “SCX’s detection rate is fundamentally optimal”)

Phase 3 (for follow-up paper, 3-6 months): 4. One clean triple-asymptotic result for a restricted setting (e.g., binary states, symmetric experts, Gaussian features)

Total Time

- **Phase 1:** 1-2 months (1 competent theoretician, or 1 month full-time)
 - **Phase 1 + 2:** 2-4 months
 - **All phases:** 6-12 months (or a dedicated PhD student over 2 years)
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9. Recommended First Step**Choose One Regime: Cluster Consistency for State Discovery**

The single most valuable and tractable asymptotic result is:

“Consistency of k-means State Discovery in SCX”

Statement: For the SCX state discovery procedure (k-means clustering on feature vectors $\phi(x_i)$), as the number of training samples $n \rightarrow \infty$ and the number of states K grows with n :

If $K = o(n^{1/3})$ and the features are regular enough (the “true” partition has well-separated components in ϕ -space), then:

$$\lim_{n \rightarrow \infty} P(\text{estimated partition} \neq \text{true partition}) = 0$$

Furthermore, if K scales as $K^* \sim C * n^{1/3}$ (for a known constant C depending on cluster geometry), the bias-variance tradeoff is optimized.

Why this is the right first step: 1. It addresses a real gap: Thm 2 only gives negative results for state discovery. A positive consistency result is long overdue. 2. It connects to a mature literature (Pollard, 1981; Von Luxburg, 2007; Rakhlin & Caponnetto, 2007) so the technical path is clear.

3. It produces a practically useful output (guidance on choosing K). 4. It can be written as a self-contained result that doesn't require simultaneous asymptotic analysis of M and n .

What the Proof Would Need

1. Assumptions:

- The “true” state partition corresponds to K well-separated components in ϕ -space (e.g., the level sets of a density, or Voronoi cells of K centroids)
- The feature map ϕ is Lipschitz continuous
- The within-state homogeneity (A5) is satisfied exactly at the true partition, with $\Delta > 0$

2. Steps:

- a. Show that k-means minimizes an empirical risk function
- b. Show that this risk converges uniformly to its population version (via uniform laws of large numbers / Glivenko-Cantelli)
- c. Show that the population risk minimizer corresponds to the true partition (identifiability)
- d. Conclude that the empirical minimizer converges to the truth
- e. Derive rates: typically $O_p(K * \log K / n)$ for the misclustering rate

3. Standard references:

- Pollard (1981, *Annals of Statistics*): “Strong consistency of k-means clustering” — the classic result for fixed K
- Rakhlin & Caponnetto (2007, *NIPS*): “Stability of k-means clustering” — covers growing K but requires well-separated clusters
- Von Luxburg (2007, *Statistics and Computing*): “A tutorial on spectral clustering” — includes consistency results

Deliverable

A 2-3 page proof (or a section in the existing Thm 2 document) showing:

1. **Theorem (SCX State Discovery Consistency):** Under conditions (sufficient feature information, well-separated true states, K growing slowly enough), the k-means state discovery procedure used in SCX recovers the true state partition with probability $\rightarrow 1$.
 2. **Corollary (Optimal K):** The number of states K^* that minimizes the expected misclassification rate scales as $O(n^{\{1/3\}})$. This is derived from balancing the approximation error (bias from using a finite K) against the estimation error (variance from finite samples per state).
 3. **Practical guidance:** For SCX practitioners: choose $K \sim n^{\{1/3\}}$ as a default, and check sensitivity by running with $K/2$ and $2K$.
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10. References

Existing SCX Theory

- Theorem 1: Multi-Expert Consistency Guarantees for Label Noise Detection (`theory/theorems/01_noise_detection_guarantee.md`)
- Theorem 2: Weak Feature Failure Lower Bound (`theory/theorems/02_weak_feature_failure_lower_bound.md`)
- Theorem 3: Unidentifiability of Noise vs. Learnable Difficulty (`theory/theorems/03_unidentifiability_theorem.md`)

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Appendix: Quick Reference — Parameter Roles in SCX

Parameter	Role	Grows -> Effect	Finite-sample result	Asymptotic target
K	Number of states	Finer partition, fewer samples per state	Thm 2: Fano bound on recovery	Optimal K* scaling
M	Number of ex-perts	More reliable consistency score	Thm 1: F1 $\exp(-M * \Delta^2)$	Minimax lower bound
n	Samples per state	More precise within-state estimates	PAC: $O(1/\sqrt{n_s})$	CLT for SCX_m(s)
eta	Noise rate	More detectable noise	Appears in F1 bound	Phase boundary
delta	Feature info	Better state recovery	Thm 2: AUC $\leq$ baseline + $O(\sqrt{\delta})$	Phase boundary

The key mathematical quantities and their asymptotic behavior:

$$SCX_m(s) = P(\text{ell}(f_m(x), y) < \tau \mid x \text{ in } s)$$

$$\text{As } n_s \rightarrow \infty: \sqrt{n_s} * (SCX_{\hat{}} - SCX) \sim N(0, SCX * (1-SCX))$$

$$C(x) = (1/M) * \sum e_m(x)$$

$$\text{As } M \rightarrow \infty: C(x) \rightarrow \mu(x) \text{ a.s.}; \quad \sqrt{M} * (C - \mu) \sim N(0, \sigma^2)$$

$$F1 \geq 1 - (1/\eta) * \exp(-2M * \Delta^2)$$

$$\text{As } M * \Delta^2 \rightarrow \infty: F1 \rightarrow 1 \text{ at exponential rate}$$

$K^* \quad (n / \log M)^{\{1/3\}} \quad [\text{conjectured optimal scaling}]$